

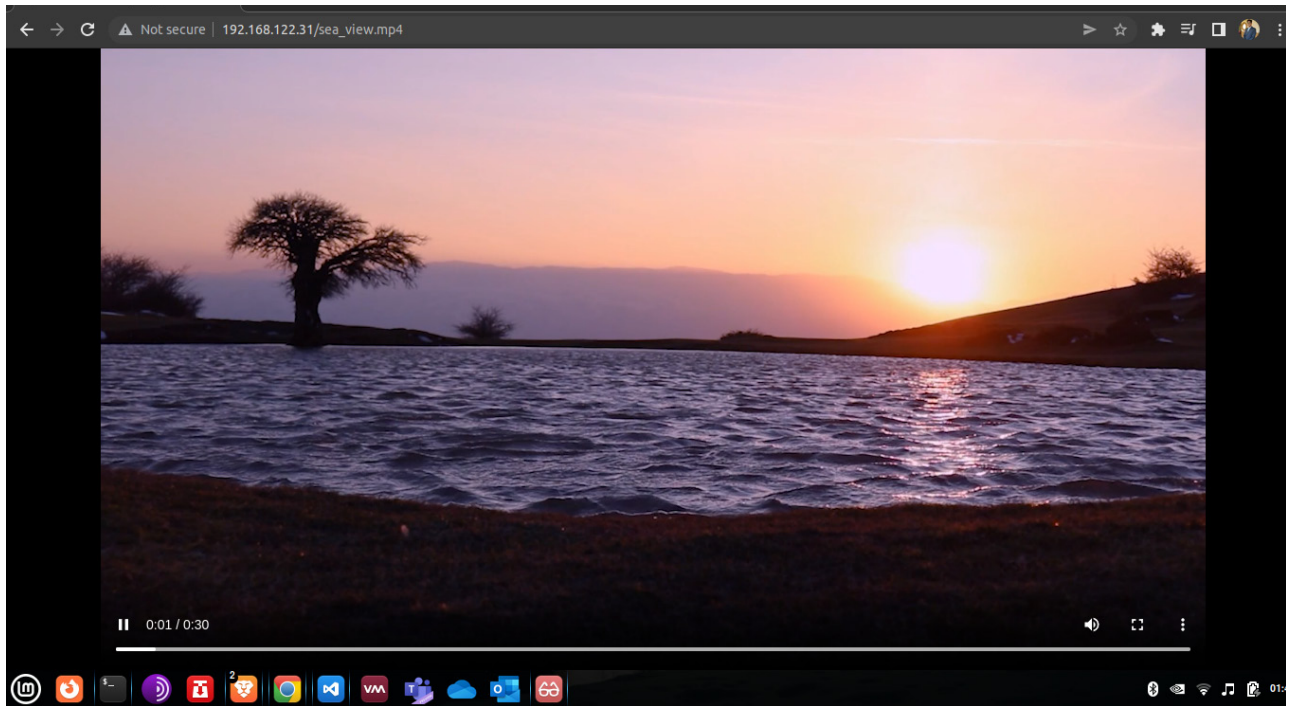


Figure 8: NFV module VOD after deployment

After the successful deployment of all configuration files and microservices over the Kubernetes cluster, the end user can access the VOD application from the web browser. Figure 8 showcases this capability. All the DevOps tools and methodologies worked in harmony to demonstrate the deployment of the NFV application. This application is deployed as a microservice and is orchestrated by Kubernetes. All the computation, network, and storage services are served by OpenStack which is configured as an Infrastructure-as-a-Service (IaaS).

Results and discussion

With the advent of DevOps tools and methodologies, we have achieved a system for NFV applications that can be deployed and delivered in a jiffy. The various methodologies of DevOps fulfil the requirement of fast-paced delivery with easy-to-configure servers and no downtime of the application. The deployed module showcases the research, where an application is created catering to the demand of video and

played over the web browser. The application can be further extended based on the requirements.

With the introduction of OpenStack to the mix of tools, the administration-based networking to the application requirements is eased out. All the networking services are facilitated via abstraction provided by OpenStack through its dashboard or APIs. OpenStack facilitates SDN, where it deploys routers, switches, load balancers, and security groups into the compute space of the virtualised application. In Figure 9, we see how each stage of the software development life cycle (SDLC) is connected via DevOps tools to produce an integrated pipeline where we successfully deploy our VOD network functions virtualization module.

Every new change in the code is populated to the GitHub and as soon as any commit is pushed, the webhook of GitHub attached to Jenkins senses the latest changes, and the job inside Jenkins gets triggered. The latest changes are pulled from the GitHub repository, which begins the process of building the code. The built code is then containerised as

a Docker image. The Docker image is pushed to the Docker registry, which is the Docker hub here. The idea behind this is that the image should be available to all the servers remotely for the active deployment of the services. In deployment, the Ansible server executes the predefined jobs of invoking the Kubernetes environment which would facilitate the NFV application. The Kubernetes infrastructure is set over the OpenStack cloud, which will encourage data confidentiality of the organisation and flourish an environment for software-defined networking.

Benefits and use cases

This is just the tip of the iceberg when it comes to the possibilities in the field of deployment of NFV applications. The experimentation successfully showcases the power and capability of DevOps tools and methodologies, where various such applications can be deployed and delivered while assuring self-healing of the deployment in case of any mishap. Kubernetes helps achieve high availability and no downtime. The rollout of new features is easily patchable to the current